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Trusting the Beep

Stated Confidence in Self-Checkout Error Prompts Does Not Predict Override-Button Use

Iben Chikere · Mirela Hanke · Casimir Beng — School of Emergent Priorities

The problem

Self-checkout kiosks flag item-recognition uncertainty with an on-screen prompt a shopper can accept or override. Retailers treat stated confidence in the prompt as a proxy for how shoppers actually use it at the till. Does professed trust move the override hand at all?

Research questions

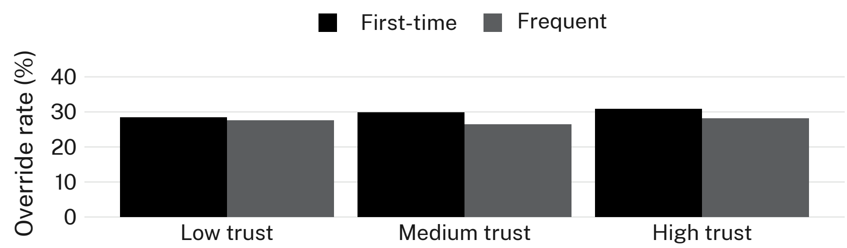
- Quantify the association between self-reported prompt trust and observed override frequency
- Test whether trust calibration (confidence matching prompt accuracy) predicts override restraint
- Establish an override-log baseline independent of shopper self-report

Study design

- $n = 238$ shoppers · 6-item trust-calibration survey · till-linked override log · 4 store sites, 12 weeks
- Mixed-effects logistic regression: override event as outcome, trust score as fixed effect, shopper and site as random effects
- Trust-tier thresholds and override definition pre-registered; no post hoc recoding

Findings

Override rate held within a narrow band (26.4-30.9%) across low, medium, and high self-reported trust tiers. The trust score explained no meaningful variance in override behaviour ($\beta = 0.04$, n.s.), and shopper frequency did not moderate the null.



Override rate ranged 26.4–30.9% across trust tiers among 238 shoppers over a 12-week override-log linkage (April–June 2026) at four store sites; first-time and frequent shoppers tracked the same flat pattern.

Interpretation

Stated trust and behavioural override move independently across trust tier, store site, and shopper tenure alike. Confidence in the prompt is not confidence in the till. Next: a randomised-wording trial testing whether prompt phrasing, not shopper attitude, is what actually moves the override hand.

Selected references

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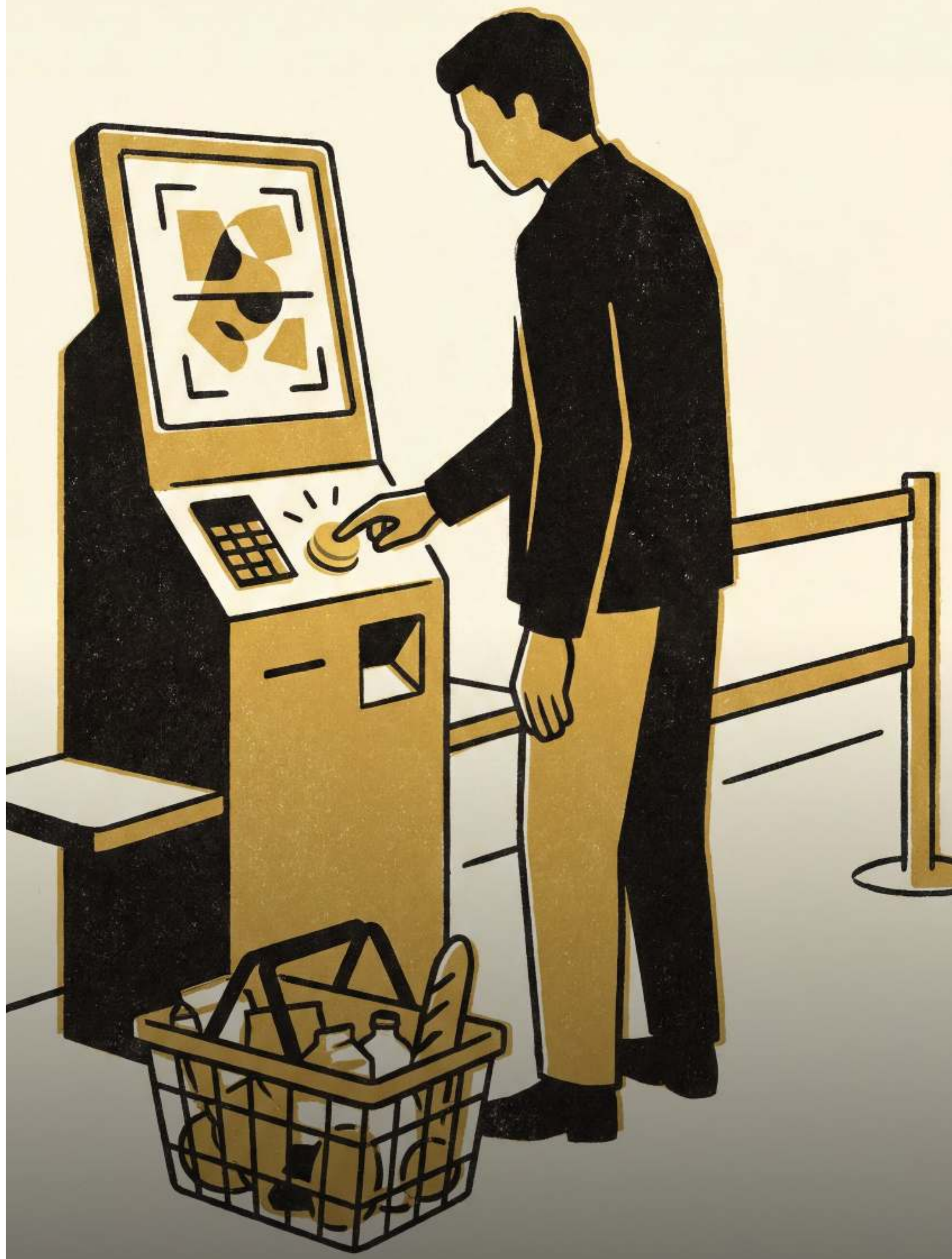
5. Tolan, A., & Beng, C. (2026). Not the kiosk's fault: An intercept-interview study of self-checkout abandonment against the library's automatic fault-ticket flag. Slop University Office of Research Outputs. doi:10.5555/slop.kwf82j

De-identified till logs only; no loyalty-card identifiers retained; trust-tier thresholds pre-registered. We thank the four participating store sites.



"The trust score and the override button never met."

Override rate held between 26% and 31% across every self-reported trust tier over a 12-week override-log linkage.